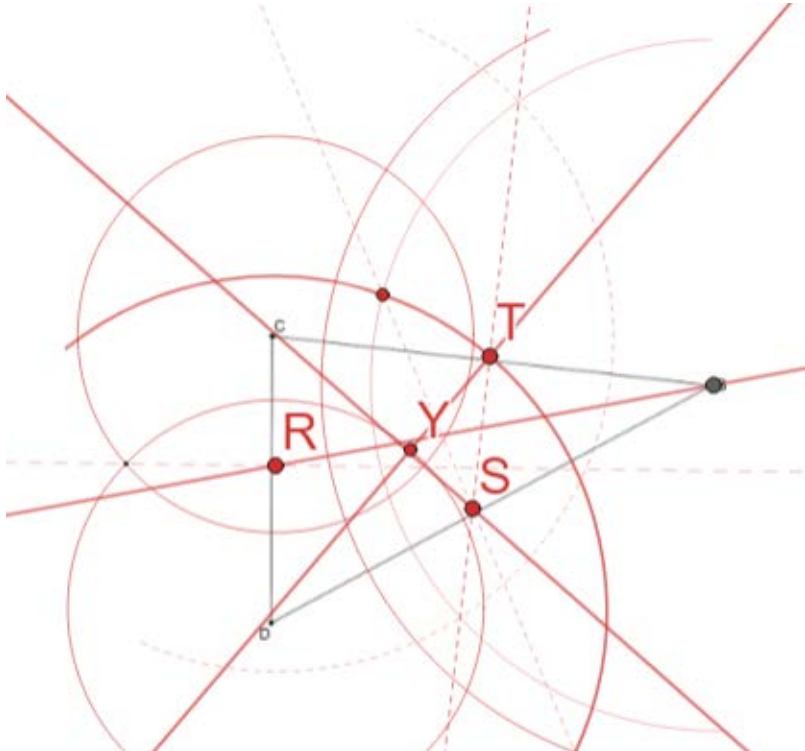
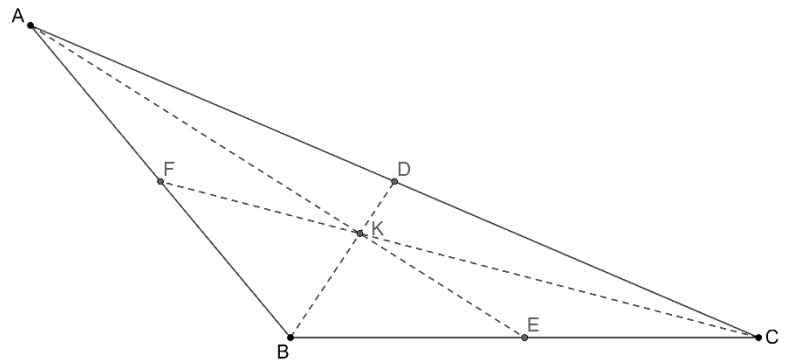


$\triangle BCD$ is shown. Complete the steps for finding the centroid of the circle. Then, label the centroid Point Y .

Steps	Construction
Construct the perpendicular bisector of BC to find its midpoint, T .	
1. Construct the perpendicular bisector of $\overline{BD}$, label midpoint S .	
2. Construct the perpendicular bisector of $\overline{CD}$, label midpoint R .	
Draw the median, R , to the opposite vertex, $\angle DBC$.	
3. Draw the median, S , to the opposite vertex, $\angle DCB$.	
4. Draw the median, T , to the opposite vertex, $\angle BDC$.	
Draw Point Y , the point of concurrency, where the three medians intersect.	

Use this information to answer questions 6 through 8.

Triangle ABC is shown, where point K is the centroid of the triangle. $FC = 6$, $BK = 2.4$, and $KE = 3.7$.



6. Find the length of $\overline{DK}$.

$$2.4 \div 2 = 1.2$$

7. Find the length of $\overline{AE}$.

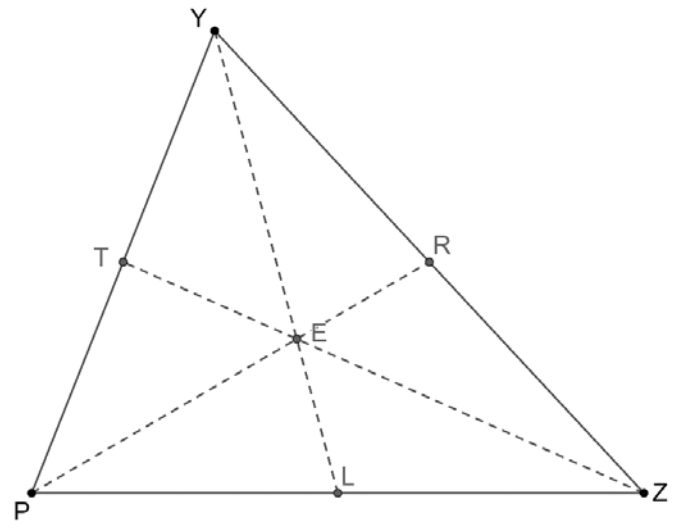
$$3.7 \cdot 3 = 11.1$$

8. Find the length of $\overline{KC}$.

$$\frac{2}{3} \cdot 6 = 4$$

Use this information to answer questions 9 and 10.

Triangle YPZ is shown, where point E is the centroid of the triangle. $TZ = 6x + 9$, and $TE = 4$.



9. Find the value of x .

$$6x + 9 = 12$$

$$6x = 3$$

$$x = \frac{1}{2}$$

10. Find the length of $\overline{EZ}$.

$$TZ = 6 \cdot \frac{1}{2} + 9 = 12$$

$$EZ = \frac{2}{3}(12) = 8$$